

Disentangled Representation Learning for Multimedia

Xin Wang

Department of Computer Science and
Technology, Tsinghua University
xin_wang@tsinghua.edu.cn

Hong Chen

Department of Computer Science and
Technology, Tsinghua University
h-chen20@mails.tsinghua.edu.cn

Wenwu Zhu

Department of Computer Science and
Technology, Tsinghua University
wwzhu@tsinghua.edu.cn

ABSTRACT

Disentangled Representation Learning (DRL) aims to learn a model capable of identifying and disentangling the underlying factors hidden in the observable data in representation form. The process of separating underlying factors of variation into variables with semantic meaning benefits in learning explainable representations of data, which imitates the meaningful understanding process of humans when observing an object or relation. As a general learning strategy, DRL has demonstrated its power in improving the model explainability, controllability, robustness, as well as generalization capacity in a wide range of scenarios such as computer vision, natural language processing, data mining etc. In this tutorial, we comprehensively present DRL from various aspects including motivations, definitions, methodologies, evaluations, applications and model designs for multimedia. We discuss works on DRL based on two well-recognized definitions, i.e., Intuitive Definition and Group Theory Definition. We further categorize the methodologies for DRL into four groups, i.e., Traditional Statistical Approaches, Variational Auto-encoder Based Approaches, Generative Adversarial Networks Based Approaches, Hierarchical Approaches and Other Approaches. We also analyze principles to design different DRL models that may benefit different tasks in practical multimedia applications. Finally, we point out challenges in DRL as well as potential research directions deserving future investigations. We believe this tutorial may provide insights for promoting the DRL research in the multimedia community.

KEYWORDS

Disentangled Representation Learning, Learning Methodology, Multimedia Applications

ACM Reference Format:

Xin Wang, Hong Chen, and Wenwu Zhu. 2023. Disentangled Representation Learning for Multimedia. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3581783.3613859>

1 TUTORIAL INFORMATION

The tutorial is planned for a half-day and organized into 5 sections.

- (1) The research and industrial motivation

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0108-5/23/10.

<https://doi.org/10.1145/3581783.3613859>

- (2) Basics and fundamentals of disentangled representation learning
- (3) Applications of disentangled representation learning
- (4) Recent advances of disentangled representation learning
- (5) Discussions and future directions

1.1 Target Audience and Prerequisites

This tutorial will be highly accessible to both the general AI community and the multimedia community, including researchers, students and practitioners who are interested in disentangled representation learning and their applications in multimedia tasks covering images, texts, videos and audio etc. The tutorial will be self-contained and designed for introductory and intermediate audiences, with an estimated audience size of 100. Although no special prerequisite knowledge is required to attend this tutorial, the audiences are supposed to have basic knowledge in machine learning, linear algebra, and calculus. In particular, audiences who have engaged in related topics (e.g., deep learning, information theory, causal inference), and multimodal generation, multimodal understanding are welcome to have Q&A interaction during the tutorial.

1.2 Motivation, Relevance and Rationale

This tutorial is to disseminate and promote the recent research achievements on disentangled representation learning as well as its multimedia applications, which is an exciting and fast-growing research direction in both the general field of machine learning and the multimodal learning. We will advocate novel, high-quality research findings, and innovative solutions to the challenging problems in disentangled representation learning, and also present how disentangled representation learning enhances different applications in multimedia. This topic is at the core of the scope of ACM Multimedia, and is attractive to ACM Multimedia audiences from both academia and industry. The objective of "Motivate and explain a topic of emerging importance for multimodal learning" will be best served by this tutorial.

2 TUTORIAL OVERVIEW

Discovering and recognizing the hidden factors behind observable data serves as one crucial step for machine learning algorithms to better understand the world. However, it still remains a challenging problem for current deep learning models which heavily rely on data representations. To solve this challenge, disentangled representation learning, as a recently cutting-edge topic in both academy and industry, aims at learning a *disentangled* representation for each object where different parts of the representation can express different (disentangled) semantics so as to improve the explainability and controllability of the machine learning models. Notably, it has achieved great success in diverse fields, such as image/video generation, recommender systems, and graph neural

networks, covering a variety of areas ranging from multimodal learning to datamining.

In this tutorial, we will disseminate and promote the recent research achievements on disentangled representation learning as well as its multimedia applications, which is an exciting and fast-growing research direction in the general field of machine learning. This tutorial consists of five parts. We first give a brief introduction on the research and industrial motivation, followed by discussions on basics, fundamentals and applications of disentangled representation learning. We will also discuss some recent advances covering disentangled learning for multimedia and disentangled graph representation learning. We finally share some of our insights on the trending for disentangled representation learning. The tutorial will be organized into the following 5 sections: i) The research and industrial motivation; ii) Basics and fundamentals of disentangled representation learning; iii) Multimedia applications of disentangled representation learning; iv) Recent advances of disentangled representation learning; v) Discussions and future directions.

3 TUTORIAL CONTENT

We introduce the most recent updates and advances in disentangled representation learning during the past years. The discussion about disentangled representation learning will be scheduled from the following three aspects: i) basics and fundamentals of disentangled representation learning, ii) multimedia application scenarios of disentangled representation learning and iii) advances of disentangled representation learning, including disentangled graph representation learning and disentangled representation for AI generated content (AIGC), etc.

3.1 Basics and Fundamentals of DRL

Current popular methods for learning disentangled representations can be roughly divided into the following categories: VAE-based, GAN-based, clustering-based methods, knowledge-guided methods.

- VAE-based methods obtain disentangled representation of an object from the perspective of probabilistic generative models. By modeling the process of the object generation with disentangled factors, this line of approaches obtain dimensionally disentangled representation.
- Different from the VAE-based methods, GAN-based methods focus on disentangling the expected factors of the object instead of disentangling each dimension of its representation.
- Clustering-based methods utilize the similarities and differences among data points to learn the disentangled concepts behind the observable data.
- Knowledge-guided methods usually give some supervisions to the model so that the designated parts of the learned representation are able to possess the expected semantics.

3.2 Applications of DRL

Disentangled representation learning easily finds its wide applications in various areas related to deep learning for its explainability and controllability. For example, in image or video generation, when we have disentangled representations for an image/video, it will be easy to generate the image/video with specific semantic attributes.

Similarly, in recommendation, when we disentangle the click behavior of users within the latent representations, the recommender systems can provide the users with not only the items but also the potential reasons why the target users may like these items, thus improving the explainability in representation learning.

3.3 Recent Advances DRL

Recently, besides the traditional image task, disentangled representation has also shown its significance in graph learning. We will first introduce some related works about disentangled relation data learning in the field like graphs and recommendation. Then, with the popularity of the recent pretrained multimodal model, we will also elaborate how disentangled representation learning relates to multimodal AIGC (AI generated content). Finally, we will give some insights on how to design disentangled strategies for your specific tasks and also some future directions.

4 ORGANIZERS

Xin Wang (<http://mn.cs.tsinghua.edu.cn/xinwang/>) is currently an Assistant Professor at the Department of Computer Science and Technology, Tsinghua University. He got both of his Ph.D. and B.E degrees in Computer Science and Technology from Zhejiang University, China. He also holds a Ph.D. degree in Computing Science from Simon Fraser University, Canada. His research interests include multimedia intelligence, machine learning and its applications in multimedia big data. He has published over 100 high-quality research papers in top journals and conferences. He is the recipient of 2017 China Postdoctoral innovative talents supporting program. He receives the ACM China Rising Star Award in 2020 and IEEE TCMC Rising Star Award in 2022.

Hong Chen (<https://forchch.github.io/>) is a Ph.D candidate at the Department of Computer Science and Technology, Tsinghua University. He received his B.E. degree at the Department of Electric and Electronic Engineering, Tsinghua University. His research interests include self-directed machine learning and multimedia intelligence. He has published several papers in top tier conferences and journals, including ICML, NeurIPS, TPAMI, ACM Multimedia, etc. He won the 1st prize of MetaDL competition in NeurIPS 2021.

Wenwu Zhu (<https://www.cs.tsinghua.edu.cn/csen/info/1161/4012.htm>) is currently a Professor in the Department of Computer Science and Technology at Tsinghua University, the Vice Dean of National Research Center for Information Science and Technology. He was a Research Manager at Microsoft Research Asia, the Chief Scientist and Director at Intel Research China and the Technical Staff at Bell Labs. He received his Ph.D. degree from New York University in 1996. His current research interests are in the area of multimedia intelligence. He received ten Best Paper Awards, including ACM Multimedia 2012 and IEEE Transactions on Circuits and Systems for Video Technology in 2001 and 2019. He served as the Editor-in-Chief for IEEE Transactions on Multimedia (2017-2019). He served in the steering committee for IEEE Transactions on Multimedia (2015-2016) and IEEE Transactions on Mobile Computing (2007-2010), respectively. He serves as General Co-Chair for ACM Multimedia 2018 and ACM CIKM 2019, respectively. He is an ACM Fellow, AAAS Fellow, IEEE Fellow, SPIE Fellow, and a member of The Academy of Europe (Academia Europaea).

REFERENCES

- [1] Hong Chen, Yudong Chen, Xin Wang, Ruobing Xie, Rui Wang, Feng Xia, and Wenwu Zhu. 2021. Curriculum Disentangled Recommendation with Noisy Multi-feedback. *Advances in Neural Information Processing Systems* 34 (2021).
- [2] Hong Chen, Yipeng Zhang, Xin Wang, Xuguang Duan, Yuwei Zhou, and Wenwu Zhu. 2023. DisenBooth: Identity-Preserving Disentangled Tuning for Subject-Driven Text-to-Image Generation. *arXiv preprint arXiv:2305.03374* (2023).
- [3] Haoyang Li, Xin Wang, Ziwei Zhang, Zehuan Yuan, Hang Li, and Wenwu Zhu. 2021. Disentangled contrastive learning on graphs. *Advances in Neural Information Processing Systems* 34 (2021), 21872–21884.
- [4] Haoyang Li, Xin Wang, and Wenwu Zhu. 2023. Curriculum Machine Learning on Graphs: A Survey. In *IJCAL*.
- [5] Haoyang Li, Ziwei Zhang, Xin Wang, and Wenwu Zhu. 2022. Disentangled graph contrastive learning with independence promotion. *IEEE Transactions on Knowledge and Data Engineering* (2022).
- [6] Jianxin Ma, Peng Cui, Kun Kuang, Xin Wang, and Wenwu Zhu. 2019. Disentangled graph convolutional networks. In *International conference on machine learning*. PMLR, 4212–4221.
- [7] Jianxin Ma, Chang Zhou, Peng Cui, Hongxia Yang, and Wenwu Zhu. 2019. Learning disentangled representations for recommendation. *Advances in neural information processing systems* 32 (2019).
- [8] Jianxin Ma, Chang Zhou, Hongxia Yang, Peng Cui, Xin Wang, and Wenwu Zhu. 2020. Disentangled self-supervision in sequential recommenders. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 483–491.
- [9] Xin Wang, Hong Chen, Si'ao Tang, Zihao Wu, and Wenwu Zhu. 2022. Disentangled Representation Learning. *arXiv preprint arXiv:2211.11695* (2022).
- [10] Xin Wang, Hong Chen, Yuwei Zhou, Jianxin Ma, and Wenwu Zhu. 2022. Disentangled Representation Learning for Recommendation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2022).
- [11] Xin Wang, Hong Chen, and Wenwu Zhu. 2021. Multimodal disentangled representation for recommendation. In *2021 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 1–6.
- [12] Xin Wang, Yudong Chen, and Wenwu Zhu. 2021. A survey on curriculum learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2021).
- [13] Xin Wang, Peng Cui, and Wenwu Zhu. 2021. Out-of-distribution Generalization and Its Applications for Multimedia. In *Proceedings of the 29th ACM International Conference on Multimedia*. 5681–5682.
- [14] Xin Wang, Xiaohan Lan, and Wenwu Zhu. 2022. Video Grounding and Its Generalization. In *Proceedings of the 30th ACM International Conference on Multimedia*. 7377–7379.
- [15] Xin Wang, Zirui Pan, Yuwei Zhou, Hong Chen, Chendi Ge, and Wenwu Zhu. 2023. Curriculum Co-disentangled Representation Learning across Multiple Environments for Social Recommendation. In *International conference on machine learning*. PMLR.
- [16] Xin Wang, Ziwei Zhang, and Wenwu Zhu. 2022. Automated graph machine learning: Approaches, libraries and directions. *arXiv preprint arXiv:2201.01288* (2022).
- [17] Xin Wang and Wenwu Zhu. 2021. Automated machine learning on graph. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. 4082–4083.
- [18] Xin Wang, Wenwu Zhu, Yonghong Tian, and Wen Gao. 2020. Multimedia Intelligence: When Multimedia Meets Artificial Intelligence. In *Proceedings of the 28th ACM International Conference on Multimedia*. 4775–4776.
- [19] Qi Wu, Peng Wang, Xin Wang, Xiaodong He, and Wenwu Zhu. 2022. *Visual Question Answering: From Theory to Application*. Springer.
- [20] Yipeng Zhang, Xin Wang, Hong Chen, and Wenwu Zhu. 2023. Adaptive Disentangled Transformer for Sequential Recommendation. In *Proceedings of the 29th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*.
- [21] Ziwei Zhang, Xin Wang, and Wenwu Zhu. 2021. Automated Machine Learning on Graphs: A Survey. In *IJCAL*.
- [22] Wenwu Zhu and Xin Wang. 2021. *Automated Machine Learning and Meta-Learning for Multimedia*. Springer.
- [23] Wenwu Zhu, Xin Wang, and Peng Cui. 2020. Deep Learning for Learning Graph Representations. In *Deep Learning: Concepts and Architectures*. Springer, 169–210.
- [24] Wenwu Zhu, Xin Wang, and Wen Gao. 2020. Multimedia Intelligence: When Multimedia Meets Artificial Intelligence. *IEEE Transactions on Multimedia* (2020).
- [25] Wenwu Zhu, Xin Wang, and Hongzhi Li. 2019. Multi-modal Deep Analysis for Multimedia. *IEEE Transactions on Circuits and Systems for Video Technology* (2019).
- [26] Wenwu Zhu, Xin Wang, and Wenpeng Zhang. 2019. AutoML and Meta-learning for Multimedia. In *Proceedings of the 27th ACM International Conference on Multimedia*. 2699–2700.